STA 640 — Causal Inference

Chapter 3.2: Propensity Score

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Central Role of Propensity score in Causal Inference

Definition (Rosenbaum and Rubin, 1983). The propensity score is defined as the conditional probability of receiving a treatment given pre-treatment covariates X:

$$e(X) = \Pr(Z = 1|X) = \mathbb{E}(Z|X), \tag{1}$$

where $X = (X_1, \ldots, X_p)$ is the collection of p covariates.

- Propensity score is a probability, analogous to a summary statistic (of the assignment mechanism)

Property 1: Balancing property

Property 1. The propensity score $e(X)$ balances the distribution of all X between the treatment groups:

$$Z \perp X \mid e(X).$$

Equivalently, $\Pr(Z_i = 1 \mid X_i, e(X_i)) = \Pr(Z_i = 1 \mid e(X_i))$.

- A balancing score $b(x)$ is a function of the covariates such that:

$$Z \perp X \mid b(X).$$

- Propensity score is a balancing score
- Rosenbaum and Rubin (1983) show that $e(X)$ is the coarsest balancing score: all balancing score is a function of $e(X)$

Remarks on the balancing property

1. If a subclass of units or a matched treatment-control pair is homogenous in $e(X)$, then the treatment and control units have the same distribution of X

2. If a subclass of units or a matched treatment-control pair is homogenous in both $e(X)$ and certain X, the other components of X within those refined class is also balanced – practical implication: estimating causal estimand in subpopulation, e.g. male or female group

3. The balancing property is a statement on the distribution of X, NOT on assignment mechanism or potential outcomes

Property 2: Unconfoundedness

Property 2. If Z is unconfounded given X, then Z is unconfounded given $e(X)$, i.e.,

$$\{Y_i(1), Y_i(0)\} \perp Z_i \mid X_i \Longrightarrow \{Y_i(1), Y_i(0)\} \perp Z_i \mid e(X_i)$$

- Given a vector of covariates that ensure unconfoundedness, adjustment for differences in propensity scores removes all biases associated with differences in the covariates
- $e(X)$ can be viewed as a summary score of the observed covariates
- Causal inference can be drawn through stratification, matching, weighting, etc. using the scalar $e(X)$ instead of the high dimensional covariates.

Propensity score: remarks

- The propensity score balances the **observed** covariates, but does not generally balance **unobserved** covariates
- In most observational studies, the propensity score $e(X)$ is unknown and thus needs to be estimated
- There is a bias-variance tradeoff between modeling $e(X)$ and directly modeling the outcome $\Pr(Y(z) \mid X)$

Propensity score analysis of causal effects

Propensity score analysis (in observational studies) typically involves two stages:

Stage 1 Estimate the propensity score $\hat{e}(X)$, e.g. by a logistic regression $(e(\boldsymbol{X}_i; \hat{\boldsymbol{\beta}}) = 1/(1 + \exp(-\boldsymbol{X}_i^T \hat{\boldsymbol{\beta}})))$ or machine learning methods.

Stage 2 Given the estimated propensity score, estimate the causal effects using:

- Stratification
- Weighting
- Matching
- Regression
- Mixed procedure of the above

Stage 1: Estimate propensity score

- The main purpose of estimating propensity score is to ensure overlap and balance of covariates between treatment groups, instead of “finding a perfect fit" or “prediction" of propensity score
- As long as the important covariates are balanced, model overfitting is usually not a concern (Rosenbaum, 1987)
- Essentially any balancing score (not necessarily propensity score) would be good enough for practical use
- A standard procedure for estimating prop score includes: initial fit, discard outliers, check covariate balance, re-fit if necessary

Stage 1: Estimate propensity score - overall flow

1. For binary treatments, most commonly via a logistic regression
2. Estimate propensity score by a logistic regression:

$$\text{logit } Pr(Z_i = 1 \mid X_i) = \beta X_i \tag{2}$$

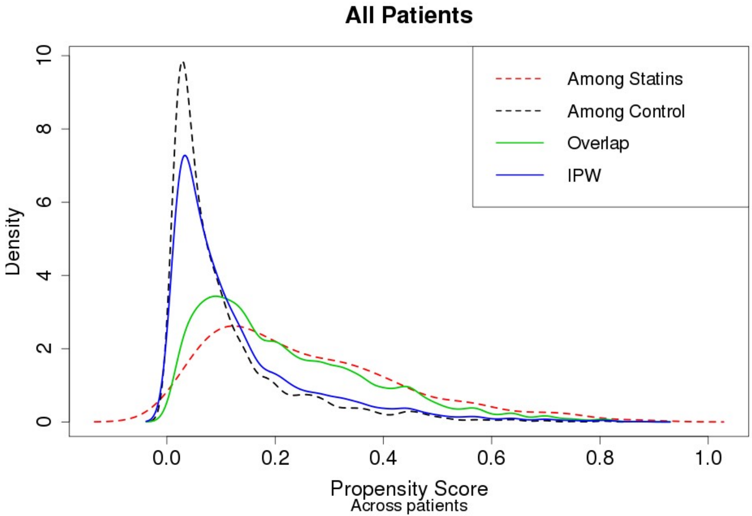
   by,e.g. a stepwise selection on the covariates and interactions) to get an initial estimate $e^0(X_i) = exp(\hat{\beta}X_i)/(1 + exp(\hat{\beta}X_i))$

3. Check overlap of propensity score between treatment groups. When the non-overlapping range is big, discard the observations with non-overlapping PS; otherwise, proceed to next step.
4. Assess balance given by initial model in (2), using matching, weighting or stratification (depends on estimation method in Stage 2)
5. If one or more covariates are seriously unbalanced: include some of their higher order terms, splines or/and interactions to re-fit the pscore model and repeat steps 2-3, until most covariates are balanced.

Stage 1: Estimate propensity score - check balance

- How to check balance depending on the estimation (of causal effects) method in Stage 2.
  - Stratification: check the balance (e.g. measured by ASD) of all important covariates within each stratum
  - Matching: check the balance of all important covariates in the matched sample
  - Weighting: check the balance of the weighted covariates between treatment and control groups.
- A useful way to visualize overlap/balance of the groups is to draw the histogram or density of the estimated PS by treatment and control groups

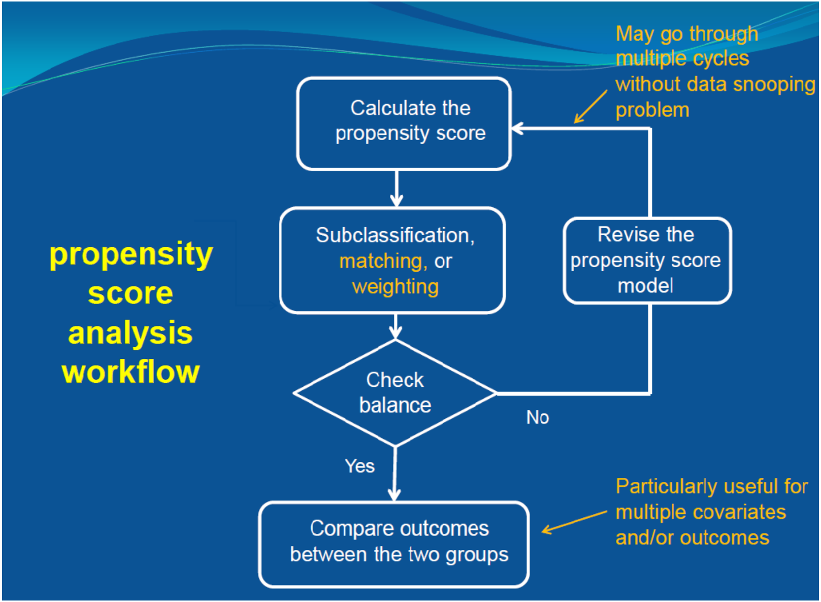
Visualization: Density Plot of PS in Framingham Study



Estimating propensity scores: flexible models

- Estimates of causal effects can be sensitive to mis-specification of propensity scores (Drake, 1993; Zhao, 2008)
- Flexible nonparametric or semiparametric models for PS:
  - splines
  - machine learning methods: CART, random forest etc.
  - Ensemble learners: generalized boosted (McCaffrey et al., 2004; R package "twang"), super learners
  - Many more...
- In low dimensions, little difference, fancy models do NOT bring advantages (Alam et al. 2019). My experience: logistic regression with some interactions are usually good enough
- In high dimensions: double machine learning – ML models for both outcome and PS (Chernozhukov et al.)

Propensity Score Analysis Workflow



What Covariates to Include in Propensity Score Model?

- What X to include int the PS model?
- First of all, X must be pre-treatment variables, conditioning on post-treatment variables induce bias (more later)
- Three types of pre-treatment covariates X
  - $\mathcal{P}$: X predictive of outcome but not trt (prognsotic covariates)
  - $\mathcal{I}$: X predictive of trt but not outcome (instrumental variables)
  - $\mathcal{S}$: X predictive of both trt and outcome (confounders)
- Include $\mathcal{S}$: deconfound
- Include $\mathcal{P}$: increase precision
- Exclude $\mathcal{I}$: including it leads to bias and loss of efficiency, known as bias amplifiers (Middleton et al. 2016), related to M-bias (Pearl)

References

Alam S., Moodie E. E. M., and Stephens D. A. (2019) Should a propensity score model be super? The utility of ensemble procedures for causal adjustment. Statistics in Medicine 38:1690-1702.

Drake, C. (1993). Effects of misspecification of the propensity score on estimators of treatment effect. Biometrics, 1231-1236.

Lunceford, J. K., Davidian, M. (2004). Stratification and weighting via the propensity score in estimation of causal treatment effects: a comparative study. Statistics in medicine, 23(19), 2937-2960.

McCaffrey, D. F., Ridgeway, G., Morral, A. R. (2004). Propensity score estimation with boosted regression for evaluating causal effects in observational studies. Psychological methods, 9(4), 403.

Rosenbaum P, Rubin D. (1983). The central role of the propensity score in observational studies for causal effects. Biometrika; 70(1):41-55.

Rosenbaum, P. R., Rubin, D. B. (1984). Reducing bias in observational studies using subclassification on the propensity score. Journal of the American statistical Association, 79(387), 516-524.

Zhao, Z. (2008). Sensitivity of propensity score methods to the specifications. Economics Letters, 98(3), 309-319.